

Scaling Safe Policy-Model Iteration with Federated Rollouts

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Situation - Configurable Environments

Most Reinforcement Learning research focuses on **static MDPs**, where the environment dynamics are fixed.

However, many real-world environments are **configurable** - certain parameters can be tuned to improve agent performance.

Configurable MDPs (Conf-MDPs) extend standard MDPs by allowing the transition model P to be selected from a space $\mathcal{P}$:

$$\mathcal{CM} = (\mathcal{S}, \mathcal{A}, R, \gamma, \mu, \mathcal{P}, \Pi)$$

- Objective: find a pair $(P, \pi) \in \mathcal{P} \times \Pi$ maximizing $J_\mu^{P, \pi}$.

Example - Car Configuration

Consider a car with **sport** and **comfort** driving modes:

- **Sport mode**: Aggressive acceleration, less stability
- **Comfort mode**: Smooth driving, better control

Policy-Configuration coupling: the best environment setting depends on the policy (and vice versa).

The goal: **jointly optimize** the policy π and environment configuration P to maximize expected return $J_\mu^{P, \pi}$.

Challenge & Solution

SPMI (Safe Policy-Model Iteration) provides **monotonic improvement guarantees** but is computationally expensive.

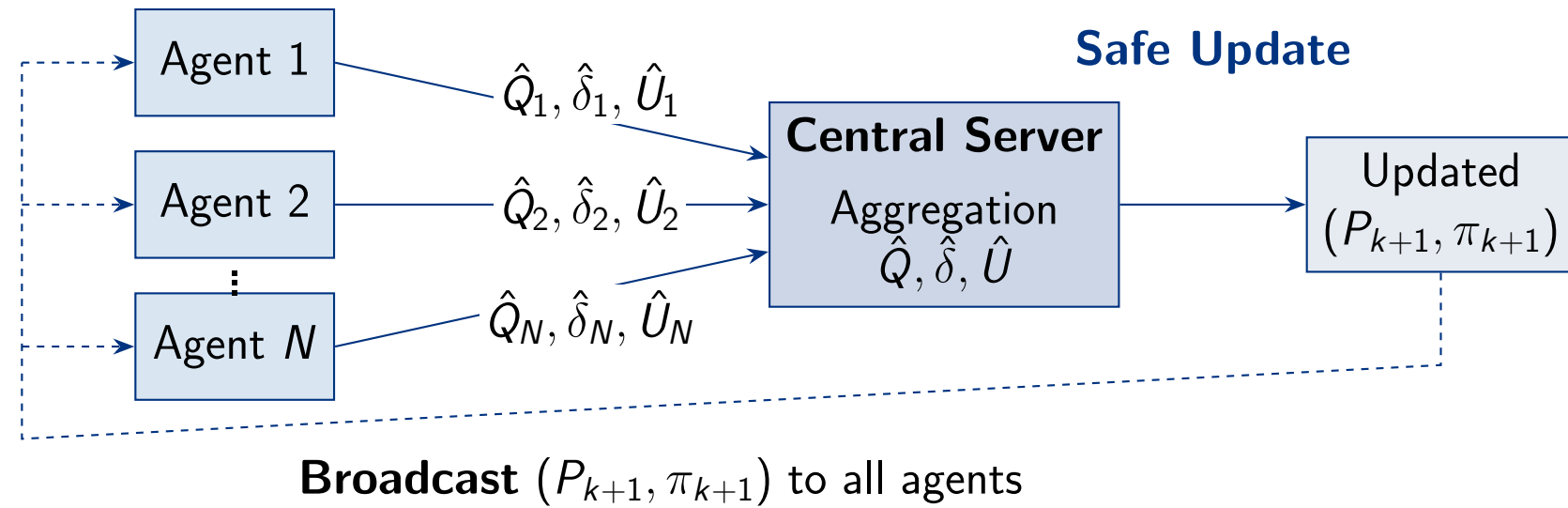
Our Solution: Federated Learning

We propose **F-SPMI** (Federated SPMI):

- Multiple agents explore in parallel
- Agents share learned value functions
- Central server aggregates knowledge
- **Safety guarantees preserved**

Faster learning through collaboration while maintaining safety!

F-SPMI Architecture



Local Statistics (per agent j):

- $\hat{Q}_j^{P_k, \pi_k}$: Q-function via Monte Carlo
- $\hat{\delta}_{\mu, j}^{P_k, \pi_k}$: γ -discounted state-action distribution
- $\hat{U}_j^{P_k, \pi_k}$: U-function estimate

Gaussian-Process-Enhanced Model Selection

Structured model search: replace a purely greedy target with a Gaussian-Process Upper Confidence Bound (GP-UCB) target selector.

$$P_{k+1} = \arg \max_{P \in \mathcal{P}} [m_k(P) + \beta_k \sigma_k(P)]$$

- $m_k(P)$ and $\sigma_k(P)$ are the GP posterior mean and uncertainty over a model-score (e.g., predicted advantage).
- Encourages exploration early (high σ_k) and exploitation later (high m_k).

Safe Update Mechanism

Weighted Federated Averaging

The server aggregates local estimates:

$$\hat{Q}^{P_k, \pi_k}(s, a) = \frac{\sum_{j=1}^N M_j(s, a) \hat{Q}_j^{P_k, \pi_k}(s, a)}{\sum_{j=1}^N M_j(s, a)}$$

where $M_j(s, a)$ is the visit count by agent j .

Decoupled Bound (Theorem 3.3, Metelli et al.)

Performance improvement is lower bounded:

$$J_\mu^{P', \pi'} - J_\mu^{P, \pi} \geq \underbrace{\frac{A_{P, \pi, \mu}^{P', \pi'} + A_{P, \pi, \mu}^{P, \pi'}}{1 - \gamma}}_{\text{advantage}} - \underbrace{\frac{\gamma \Delta Q^{P, \pi} D}{2(1 - \gamma)^2}}_{\text{penalization}} = \hat{B}(P', \pi')$$

Safe Step Sizes

Optimal update coefficients:

$$\alpha^*, \beta^* = \arg \max_{\alpha, \beta} \hat{B}(\alpha, \beta)$$

ensuring **monotonic performance improvement**. I.e., we optimize coefficients that in turn are used for convex mixing updates.

Convex mixing update: $\pi_{k+1} = \alpha^* \pi_{\text{tgt}} + (1 - \alpha^*) \pi_k$, $P_{k+1} = \beta^* P_{\text{tgt}} + (1 - \beta^*) P_k$.

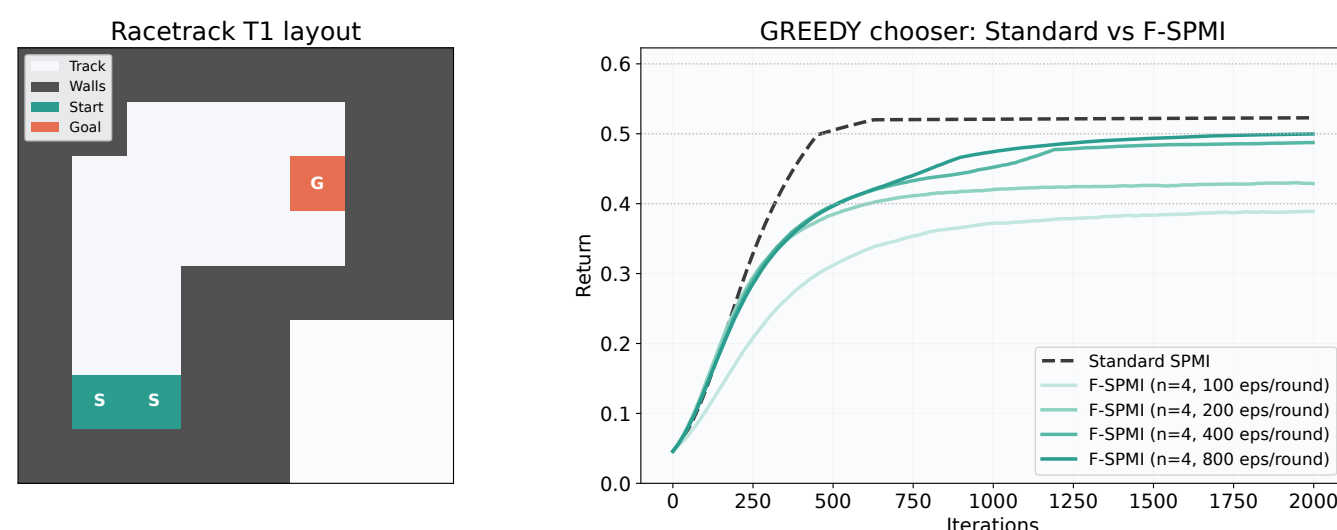
Experimental Setup & Sample Efficiency

Domain: Racetrack-T1 Conf-MDP.

Metric: Discounted performance $J_\mu^{P, \pi} = \mu^\top V^{P, \pi}$ via exact evaluator.

Observation: More episodes per round yields a closer match to SPMI; with fewer episodes, higher variance slows improvement.

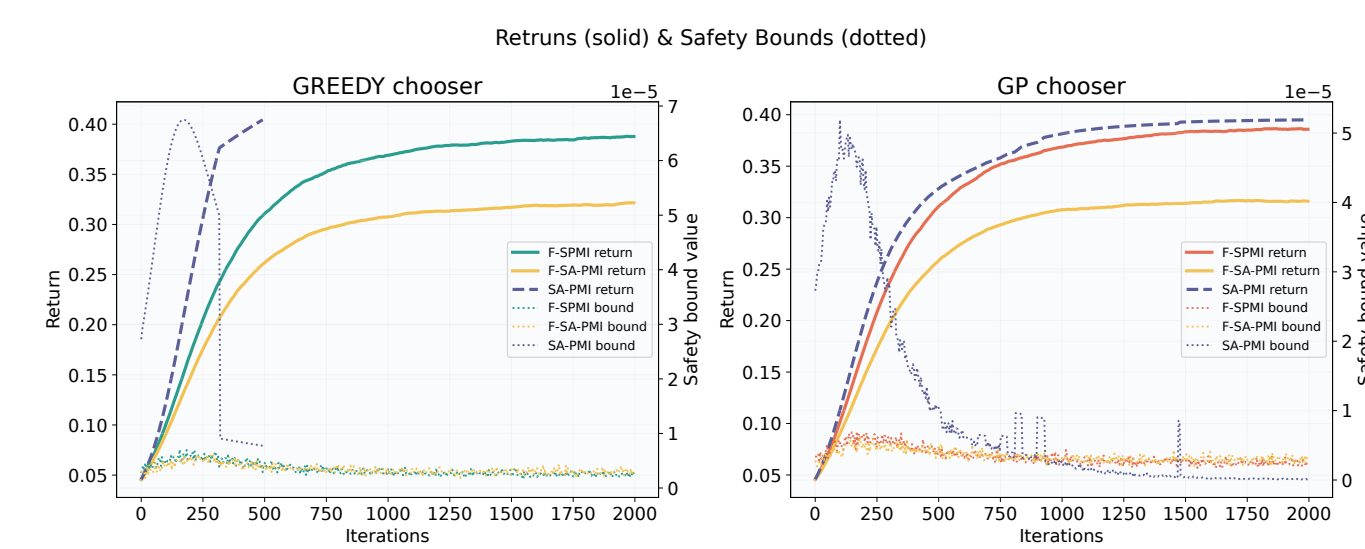
Experimental setup & sample efficiency (varying episodes per round)



Convergence & Safety Guarantees

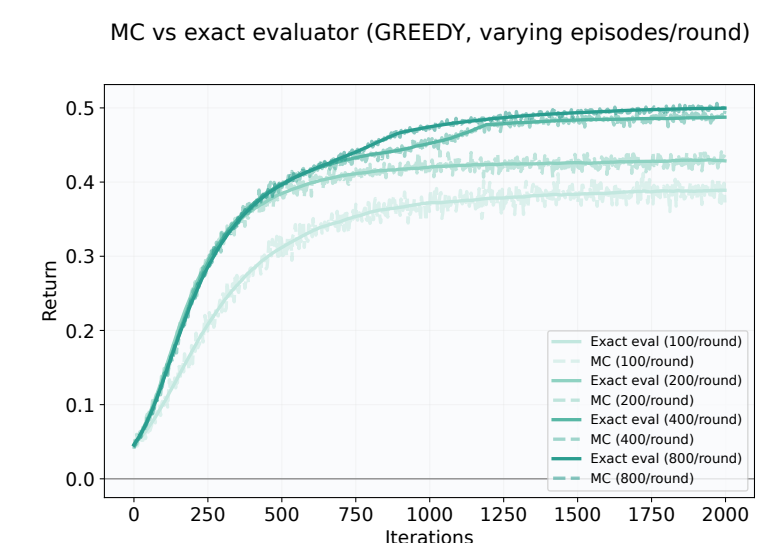
Safety signal: the dotted curves show the estimated guaranteed-improvement bound $\hat{B}$. The bound tightens over time as updates shrink, consistent with convergence of step sizes.

Conclusion: In this finite, tabular setting, GP-based target selection matches/slightly underperforms greedy; we *hypothesize* a greater impact in larger environments or structured model spaces.



MC Estimate vs. Exact Evaluator

Agreement: For the same learned (P, π) , MC estimates of $J_\mu^{P, \pi}$ closely match the *exact evaluator*; with fewer episodes, variance increases and estimates are noisier.



Implication: MC estimation enables practical deployment where exact evaluation is computationally prohibitive, scaling to larger state-action spaces without matrix inversions

Summary

Key Contributions:

- Federated SPMI executes **policy-model updates in parallel** while preserving safe improvement guarantees
- **Decoupled bound** yields monotonic policy/ model refinement from finite-sample statistics
- Aggregation is **robust to stochasticity/ heterogeneity**; MC returns track exact performance
- Optional **GP-UCB** target selection exploits structure of the model space

Limitations

Current Constraints:

- Federation uses MC estimates with finite samples - variance affects the confidence bound
- Assumes a **known finite model set** $\mathcal{P}$ and tabular state-action space
- **Synchronous** rounds; all agents must participate each update
- Communication cost scales with number of agents and statistics exchanged
- Safety guarantees impose limitations on performance and convergence

Outlook

Future Directions:

- Learn or approximate $\mathcal{P}$ from data; extend to latent/ continuous model parametrizations
- **Function approximation** and policy-gradient variants for large/ continuous spaces with safe steps
- **Asynchronous** or partial-participation protocols robust to malicious agents
- Deployment on robotic/autonomous platforms under bandwidth and safety constraints

In Short: Federated rollouts boost sample throughput for safe updates. Monte Carlo evaluation avoids exact value computation and, with moderate episodes per round, closely matches the *exact evaluator* - potentially supporting scalable learning in larger Conf-MDPs.